Cross-Layer State Consistency in Consumer Remote SIM Provisioning*

Jhury Kevin Lastre[1] and Ilsun You[1,2]†

[1] Kookmin University, Seoul, South Korea
{lavelliane,isyou}@kookmin.ac.kr
[2] KMU Global Research Center for ICT Convergence Security, Kookmin University, Seoul, South Korea

Abstract

Consumer Remote SIM Provisioning (RSP) enables digital mobile operators, travel connectivity platforms, and device-based subscription services to sell and activate mobile connectivity without physical SIM distribution. However, a Consumer RSP transaction is not only a technical profile download. It is a business workflow involving order creation, payment confirmation, activation-code delivery, profile download, customer entitlement, billing activation, service suspension, refund, profile deletion, and subscription revocation. This paper presents a practical methodology for testing whether Consumer RSP business processes remain consistent with technical profile state and network access. We build an open testbed using eqSIM `smdpp-c`, lpac, a sysmoEUICC-C2T card, Open5GS, and UERANSIM to model the lifecycle of a Consumer RSP service from purchase to network admission. We define cross-layer invariants that connect business state, SM-DP+ state, eUICC profile state, notification status, mobile-core subscriber state, and registration outcome, and we model-check them in TLA+. We study five failure cases: reused activation codes, failed downloads after payment, deleted profiles with active billing, revoked subscriptions that still receive network access, and stale customer entitlements after profile deletion. We propose receipt-based reconciliation as a lightweight mechanism for aligning provisioning, billing, customer support, and network-control systems, and evaluate it against a no-reconciliation baseline on the live testbed. Reconciliation detects all five cases and, for the four repairable ones, restores consistency within sub-second exposure windows where the baseline leaves them indefinitely open. This work provides a systematic foundation for evaluating Consumer RSP platforms as end-to-end digital service businesses, not only as telecom provisioning systems.

Acknowledgment. This work was supported by the Institute for Information & Communications Technology Promotion (IITP) grant funded by the Korea government (MSIT) (No. RS-2024-00437252, Development of anti-sniffing technology for mobile communication and AirGap environments).

*Proceedings of The 2026 IFIP WG 8.4 International Symposium on E-Business Information Systems Evolution (EBISION 2026), Article No. P-6, July 8–10, 2026, Seoul, South Korea.
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†Corresponding author